Document 10: Implementation Roadmap & Development Phases

Project: Natural Intellects School Management System (NISMS)

Version: 1.0

Status: Draft

Purpose:

This document defines the step-by-step execution plan for building, testing, and deploying the Natural Intellects School Management System.

It converts all prior documentation into actionable development phases.

It serves as the master execution guide for:

- Cloud AI development
- Human developers
- Figma designers
- Database engineers
- Deployment teams
- Testing teams

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1. IMPLEMENTATION PHILOSOPHY

The system shall be built in a structured, layered approach:

No Parallel Chaos

Phases must be completed sequentially to avoid:

- Rework
- Inconsistencies
- Misaligned designs
- Broken integrations

Single Source of Truth

Each phase must strictly follow:

- Document 04 (FRS)
- Document 05 (Workflows)
- Document 06 (Architecture)
- Document 07 (Database)
- Document 08 (UI/UX)
- Document 09 (Screens)

AI-Assisted Development

Cloud AI shall be used for:

- Code generation
- Schema creation

- API scaffolding
- UI generation

But always constrained by documentation.

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2. PHASE OVERVIEW

Phase Flow

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```
Validation → Design → Database → Backend → Frontend → Integration → Testing
→ Pilot → Production → Training
```

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3. PHASE 1: SYSTEM FINALIZATION & VALIDATION

Objective

Ensure all documentation is complete and consistent.

Tasks

- Review all documents (04–09)
- Remove contradictions
- Confirm feature scope
- Validate workflows
- Freeze feature set

Output

- Locked Product Specification
- Final Scope Agreement

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4. PHASE 2: FIGMA DESIGN & PROTOTYPING

Objective

Translate Document 08 & 09 into visual prototypes.

Tasks

- Build design system
- Create wireframes
- Build full UI screens
- Prototype workflows
- Mobile responsiveness testing

Output

- Complete Figma project
- Clickable prototype
- Design system library

5. PHASE 3: DATABASE IMPLEMENTATION

Objective

Implement PostgreSQL schema from Document 07.

Tasks

- Create Prisma schema
- Define relationships
- Implement migrations
- Seed initial data
- Validate constraints

Output

- Working database
- Migration system
- Seeded environment

6. PHASE 4: BACKEND DEVELOPMENT

Objective

Build system logic and APIs.

Tasks

- Authentication system
- Role-based access control
- Student APIs
- Teacher APIs
- Finance APIs
- Academic APIs
- CMS APIs

Output

- Functional backend
- API endpoints
- Secure access control

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7. PHASE 5: FRONTEND DEVELOPMENT

Objective

Build UI based on Figma designs.

Tasks

- Public website
- Admin dashboard
- Teacher portal
- Bursar portal
- Mobile responsiveness

Output

- Fully functional frontend
- Connected UI components
- Responsive system

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8. PHASE 6: INTEGRATION PHASE

Objective

Connect frontend, backend, and database.

Tasks

- API integration
- Data binding
- Authentication linking
- File storage integration
- Role-based routing

Output

- Fully connected system
- Working end-to-end flows

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9. PHASE 7: TESTING & QUALITY ASSURANCE

Objective

Ensure system stability and correctness.

Testing Types

- Unit testing
- Integration testing
- UI testing
- Security testing
- Performance testing

Output

- Bug reports
- Fixed system
- Stability confirmation

10. PHASE 8: PILOT DEPLOYMENT

Objective

Deploy system to Natural Intellects pilot school.

Tasks

- Deploy staging environment
- Onboard real users
- Monitor usage
- Collect feedback

Output

- Real-world validation
- User feedback data

11. PHASE 9: PRODUCTION DEPLOYMENT

Objective

Launch system publicly.

Tasks

- Final deployment
- Domain configuration
- SSL setup
- Performance optimization

Output

- Live system
- Public access

12. PHASE 10: TRAINING & ADOPTION

Objective

Ensure users can operate the system.

Tasks

- Staff training
- User manuals
- Video guides
- Onboarding sessions

Output

- Trained school staff
- Adoption readiness

13. PHASE 11: MAINTENANCE & ITERATION

Objective

Long-term system sustainability.

Tasks

- Bug fixes
- Feature improvements
- Performance monitoring
- Security updates

Output

- Stable long-term system
- Iterative improvements

14. RISK MANAGEMENT

Technical Risks

- Integration failures
- Database scaling issues
- API breakdowns

Operational Risks

- Low user adoption
- Training gaps
- Internet limitations

Mitigation

- Strong documentation
- Pilot testing
- Incremental rollout

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15. EXECUTION PRINCIPLES

Discipline Over Speed

Proper structure matters more than fast delivery.

Documentation First

No code shall be written without reference to:

- FRS
- Workflows
- Architecture
- Database design
- UI/UX specification

Iterative Improvement

Each phase improves clarity for the next.

STRATEGIC CONCLUSION

This roadmap ensures that the Natural Intellects School Management System is:

- Predictable in development
- Structured in execution
- Scalable in architecture
- Consistent in design
- Ready for real-world deployment

End of Document 10.